

LAB 13

(Total Marks 10)

Task 1: Create a program that utilizes interrupts to detect and display whether the keys (all alphabetic in your name) are pressed on the keyboard. The program should continue running until the Esc key is pressed, at which point it should exit the loop. (Hint int 16h)

```
org 0x0100
jmp start
m: db 'm is pressed', 0
u: db 'u is pressed', 0
h: db 'h is pressed', 0
a: db 'a is pressed', 0
d: db 'd is pressed', 0
t: db 't is pressed', 0

M: db 'M is pressed', 0
U: db 'U is pressed', 0
H: db 'H is pressed', 0
A: db 'A is pressed', 0
D: db 'D is pressed', 0
T: db 'T is pressed', 0
notFound: db 'Wrong char presed', 0
escPressed: db 'ESC presed', 0

clrscr:
    push ax
    push es
    push di
    push cx

    mov ax, 0xb800
    mov es, ax
    mov di, 0
    mov ax, 0x0720
    mov cx, 2000

    cld
    rep stosw

    pop cx
    pop di
    pop es
    pop ax
    ret

strlen:
    push bp
    mov bp, sp
    push es
    push cx
    push di

    les di, [bp+4] ; point es:di to string
```

```

mov cx, 0xffff
xor al, al
repne scasb

mov ax, 0xffff
sub ax, cx
dec ax ; ;str length

pop di
pop cx
pop es
pop bp
ret 4

```

printstr:

```

push bp
mov bp, sp
push ax
push es
push di
push si
push cx
push ds

push ds
mov ax, [bp+4]
push ax
call strlen
mov cx, ax          ;length

cmp cx, 0
jz exit

mov al, 80
mul byte [bp+8] ;y value
add ax, [bp+10] ;x value
shl ax, 1
mov di, ax

mov ax, 0xb800
mov es, ax
mov si, [bp+4] ;si point to string

xor ax, ax
mov ah, [bp+6] ;attribute

```

cld

l1:

```

lodsb
stosw
loop l1

```

exit:

```

pop ds
pop cx
pop si
pop di
pop es

```

```
pop ax
pop bp
ret 8
```

checkchar:

```
push bp
mov bp, sp
push ax
push bx
push dx
```

```
mov dx, 2 ;green attribute if correct char pressed, it is changed below if wrong input is given
```

```
;;;;;;;;;;Capital letter input;;;;;;;;;;
```

```
cmp al,'M'
je printM
```

```
cmp al,'U'
je printU
```

```
cmp al,'H'
je printH
```

```
cmp al,'A'
je printA
```

```
cmp al,'D'
je printD
```

```
cmp al,'T'
je printT
```

```
;;;;;;;;;;Small letter input;;;;;;;;;;
```

```
cmp al,'m'
je printm
```

```
cmp al,'u'
je printu
```

```
cmp al,'h'
je printh
```

```
cmp al,'a'
je printa
```

```
cmp al,'d'
je printd
```

```
cmp al,'t'
je printt
```

```
cmp al, 0x1B
je escPrint
```

```
jne noFound
```

```
;;;;;;;;;;Capital letter input;;;;;;;;;;
```

printM:

```

        mov bx, M
        jmp print
printU:
        mov bx, U
        jmp print
printH:
        mov bx, H
        jmp print
printA:
        mov bx, A
        jmp print
printD:
        mov bx, D
        jmp print
printT:
        mov bx, T
        jmp print

        ;;;;;;;;;Small letter input;;;;;;;;;
printm:
        mov bx, m
        jmp print
printu:
        mov bx, u
        jmp print
printh:
        mov bx, h
        jmp print
printa:
        mov bx, a
        jmp print
printd:
        mov bx, d
        jmp print
printt:
        mov bx, t
        jmp print

noFound:
        mov dx, 1                ;blue attribute if wrong char pressed
        mov bx, notFound
        jmp print

escPrint:
        mov dx, 4                ;red attribute if ESC is pressed
        mov bx, escPressed

print:
        mov ax, 6                ;x value
        push ax

        mov ax, [bp+4] ;y value
        push ax

        push dx                ; push attribute

        push bx                ;string offset

```

```
call printstr
```

```
pop dx  
pop bx  
pop ax  
pop bp  
ret 2
```

start:

```
call clrscr ; clear the screen
```

```
mov cx, 2 ; y value
```

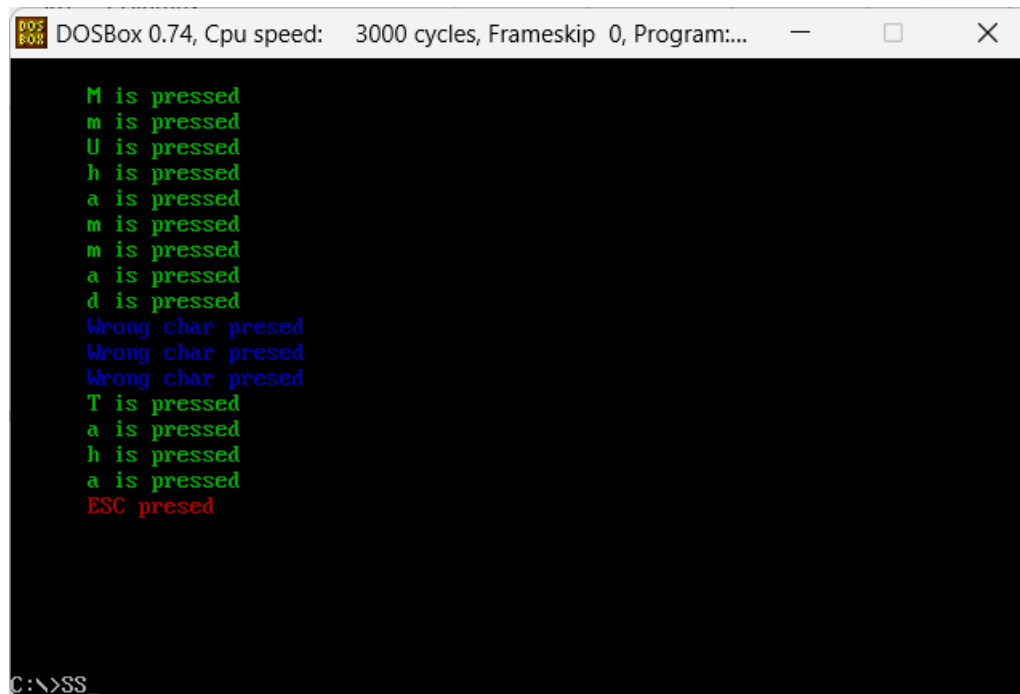
l2:

```
;taking input  
mov ah, 0  
int 0x16  
mov ah, 0
```

```
push cx  
call checkchar  
inc cx ; inc cx (y-value) to print on next line
```

```
cmp al, 0x1B ;checking ESC as input  
jne l2
```

```
mov ax, 0x4c00 ; terminate program  
int 0x21
```



```
DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program:...  
  
M is pressed  
m is pressed  
U is pressed  
h is pressed  
a is pressed  
m is pressed  
m is pressed  
a is pressed  
d is pressed  
Wrong char presed  
Wrong char presed  
Wrong char presed  
T is pressed  
a is pressed  
h is pressed  
a is pressed  
ESC presed  
  
C:\>SS_
```

Task 2: Write a program that prompts the user to enter two integers within range 5 and then display their sum on the screen

```
org 0x0100
jmp start
num1: db 'Enter Num 1', 0
num2: db 'Enter Num 2', 0
num: db 0
sumMsg: db 'Sum is', 0
escPressed: db 'ESC pressed', 0
wrong: db 'Wrong Input, Press key to Retry or ESC to quit', 0
```

clrscr:

```
    push ax
    push es
    push di
    push cx

    mov ax, 0xb800
    mov es, ax
    mov di, 0
    mov ax, 0x0720
    mov cx, 2000

    cld
    rep stosw

    pop cx
    pop di
    pop es
    pop ax
    ret
```

strlen:

```
    push bp
    mov bp, sp
    push es
    push cx
    push di

    les di, [bp+4] ; point es:di to string
    mov cx, 0xffff
    xor al, al
    repne scasb

    mov ax, 0xffff
    sub ax, cx
    dec ax ; ;str length

    pop di
    pop cx
    pop es
    pop bp
    ret 4
```

printstr:

```
    push bp
```

```
mov bp, sp
push ax
push es
push di
push si
push cx
push ds
```

```
push ds
mov ax, [bp+4]
push ax
call strlen
mov cx, ax          ;length
```

```
cmp cx, 0
jz exit
```

```
mov al, 80
mul byte [bp+8] ;y value
add ax, [bp+10] ;x value
shl ax,1
mov di, ax
```

```
mov ax, 0xb800
mov es, ax
mov si, [bp+4] ;si point to string
```

```
xor ax, ax
mov ah, [bp+6] ;attribute
```

```
cld
```

l1:

```
lodsb
stosw
loop l1
```

exit:

```
pop ds
pop cx
pop si
pop di
pop es
pop ax
pop bp
ret 8
```

printNum:

```
push bp
mov bp, sp
push bx
push ax
push di
push es
```

```
mov bx, 0xb800
mov es, bx
add al, 0x30
mov ah, 1
```

```
mov di, [bp+4]
mov word [es:di], ax ; print char on screen
```

```
pop es
pop di
pop ax
pop bx
pop bp
ret 2
```

sum:

```
push bp
mov bp, sp
push bx
push ax
```

print:

```
mov ax, 6      ;x value
push ax
```

```
mov ax, 7      ;y value
push ax
```

```
mov ax, 3
push ax        ; push attribute
```

```
mov ax, sumMsg
push ax        ;string offset
```

```
call printstr
```

```
;calculating sum
add bh, bl
mov byte [num], bh      ;sum is stored in num
```

```
mov al, bh
mov di, 1448
push di
call printNum
```

```
pop ax
pop bx
pop bp
ret 2
```

start:

```
mov cx, 2
```

l2:

```
;;;;;;;;;Print MSG for NUM1 input;;;;;;;;;
call clrscr ; clear the screen
mov ax, 2      ;x value
push ax
```

```
mov ax, 2      ;y value
push ax
```



```

mov ax, 1
push ax                ; push attribute

mov ax, num1 ;string offset
push ax
call printstr

; ;;;;;;;;;;taking NUM 1 input;;;;;;;;;;

mov ah, 0
int 0x16
sub al,0x30

cmp al, 4
jnbe wrongInput        ;jmp if not below of equal

xor bx,bx
mov bl,al               ;storing num1 in bl

mov di, 484
push di
call printNum

;;;;;;;;;;Print MSG for NUM2 input;;;;;;;;;;
mov ax, 2                ;x value
push ax

mov ax, 5                ;y value
push ax

mov ax, 1
push ax                ; push attribute

mov ax, num2 ;string offset
push ax
call printstr

;;;;;;;;;;taking NUM 2 input;;;;;;;;;;
mov ah, 0
int 0x16
mov ah, 0
sub al,0x30

cmp al, 4
jnbe wrongInput        ;jmp if not below of equal

mov bh, al              ;storing num2 in bh

mov di, 966
push di
call printNum

;calculate sum
push bx
call sum

```

jmp quit ;exit after calculating correct sum

wrongInput:

mov ax, 2 ;x value
push ax

mov ax, 6 ;y value
push ax

mov ax, 4
push ax ; push attribute

cmp al, 0x1B
je printESC ;if ESC was pressed as Wrong Input than it prints ESC and exits

mov ax, wrong ;string offset
push ax
call printstr

mov ah, 0
int 0x16 ;input to continue

cmp al, 0x1B ;checking if again ESC as input
jne l2

printESC:

mov ax, escPressed
push ax
call printstr

quit:

mov ax, 0x4c00 ; terminate program
int 0x21



Best of luck